

## CASE LETTERS

**Rebound erythema and burning sensation from a new topical brimonidine tartrate gel 0.33%**

*To the Editor:* Brimonidine tartrate (BT) gel 0.33% (Mirvaso, Galderma Laboratories LP), an alpha-2 selective adrenergic receptor agonist and vasoconstrictor, reduces erythema of rosacea.<sup>1-3</sup> The top 3 reported adverse events related to BT are erythema worse than baseline (4%), flushing (3%), and burning (2%). At least 1% of patients had an adverse reaction to the medication.<sup>1,3</sup>

We report severe erythema in conjunction with burning sensation in 3 patients using BT. All reported reduced erythema 1 to 6 hours after application, followed by self-limited erythema, more severe than baseline, accompanied by burning over the next 12 hours and lasting roughly 12 hours. This phenomenon worsens with continued use. Urticaria and facial edema were absent. Even though the Mirvaso package insert reports infrequent erythema worse than baseline, this number may be higher in practice.<sup>1,2</sup>

**Patient 1.** A 36-year-old Latina woman with long-standing rosacea and Hashimoto thyroiditis applied BT to cheeks and nose as directed. Erythema reduction lasting 4 hours was followed by erythema, notably elevated over baseline, and burning in the areas where the BT was applied (Fig 1). This reaction resolved after 5 hours. The patient continued to use the BT over the course of 5 days, applying in 5 areas, as directed, to her forehead, each cheek, nose, and chin. She felt the reduction in erythema was satisfactory but noted that patchy erythema and burning developed 5 to 10 hours after application. On the 5th day of use, 12 hours after application, she experienced erythema, much worse than at baseline, in all areas where the medication had been applied and she had a severe skin burning sensation (Fig 2). The erythema and burning resolved 12 hours later and she discontinued use of BT.

**Patient 2.** A 41-year-old Caucasian woman with long-standing rosacea, psoriasis, and anxiety reported reduced erythema following one application of BT, followed 3 hours later by significant erythema, skin burning, and pruritus where BT was applied. These resolved slowly over the next 12 hours. Concurrent medications included infliximab, methotrexate, folic acid, venlafaxine, zolpidem, clonazepam, and trazodone.

**Patient 3.** A 28-year-old Caucasian woman with long-standing rosacea and Hashimoto thyroiditis,

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**Fig 1.** Rosacea, patient 1. Four hours after application of brimonidine tartrate to cheeks and nose. Note that erythema is absent from her forehead, where the medication was not applied.

taking levothyroxine, tramadol, and norethindrone/ethinyl estradiol, applied BT to cheeks and chin once daily for 2 weeks. She noted significant reduction in erythema followed 6 to 8 hours later by erythema and burning in the areas of application. She continued therapy for 2 weeks with progressively worsening rebound erythema and burning. These symptoms resolved temporarily with reapplication of BT.

There is no current photo documentation of this side effect at [www.mirvaso.com](http://www.mirvaso.com) or in the literature.<sup>3</sup> We propose that this reaction constitutes rebound dilation of the capillaries caused by down-regulation of alpha-2 adrenergic receptors following use of BT. This may be similar to rhinitis medicamentosa, observed with overuse of



**Fig 2.** Rosacea, patient 1. Twelve hours after application of brimonidine tartrate to forehead, cheeks, nose, and chin. Erythema is present in all of the areas where the medication was applied.

alpha-adrenergic agonist nasal sprays (eg, oxymetazoline and xylometazoline).<sup>4</sup> This reaction directly opposes the goal of therapy.

Counseling about the potential for worsening erythema, use of a test area, and limiting use to special occasions may be warranted. Viewing images of this adverse reaction prior to use may alleviate patient distress.

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## Acne fulminans successfully treated with cyclosporine and isotretinoin

*To the Editor:* Acne fulminans (AF) is a rare and very severe form of acne characterized by the sudden onset of painful, ulcerative pustules and associated systemic symptoms. Recommended treatment is a combination of oral steroids and isotretinoin. We report the case of a male patient with poor response to this treatment who was successfully treated with the combination of oral cyclosporine and isotretinoin.

A 15-year-old white male patient with severe acne flare was referred to our hospital. Diagnosis was acne vulgaris for the past 2 years, treated with topical antibiotics. Six months before the consultation, the lesions worsened and a diagnosis of acne conglobata was made. He was treated with isotretinoin 20 mg/day and prednisone 15 mg/day, but there was no clinical improvement. Isotretinoin dosage was increased to 30 mg/day, but 3 weeks later multiple reddish papulonodular and ulcerated lesions with hemorrhagic crusts suddenly developed on his face, neck, and trunk. The lesions were painful, and arthralgias and temperature up to 39°C were noted (Fig 1). Abnormal laboratory findings included elevated C-reactive protein levels (5 mg/dL; normal < 1 mg/dL) and leukocytosis (15,700 cells/mm<sup>3</sup>) with neutrophilia (68.8%). AF was diagnosed, and treatment with prednisone 60 mg/day and isotretinoin 20 mg/day was initially successful. Nevertheless progressive worsening was observed in the following weeks while prednisone was tapered and isotretinoin increased to 30 mg/day. He was treated with potassium permanganate baths and topical antibiotics. Oral cyclosporine 5 mg/kg/day plus isotretinoin 30 mg/day was initiated and systemic steroids were stopped. After a few weeks,